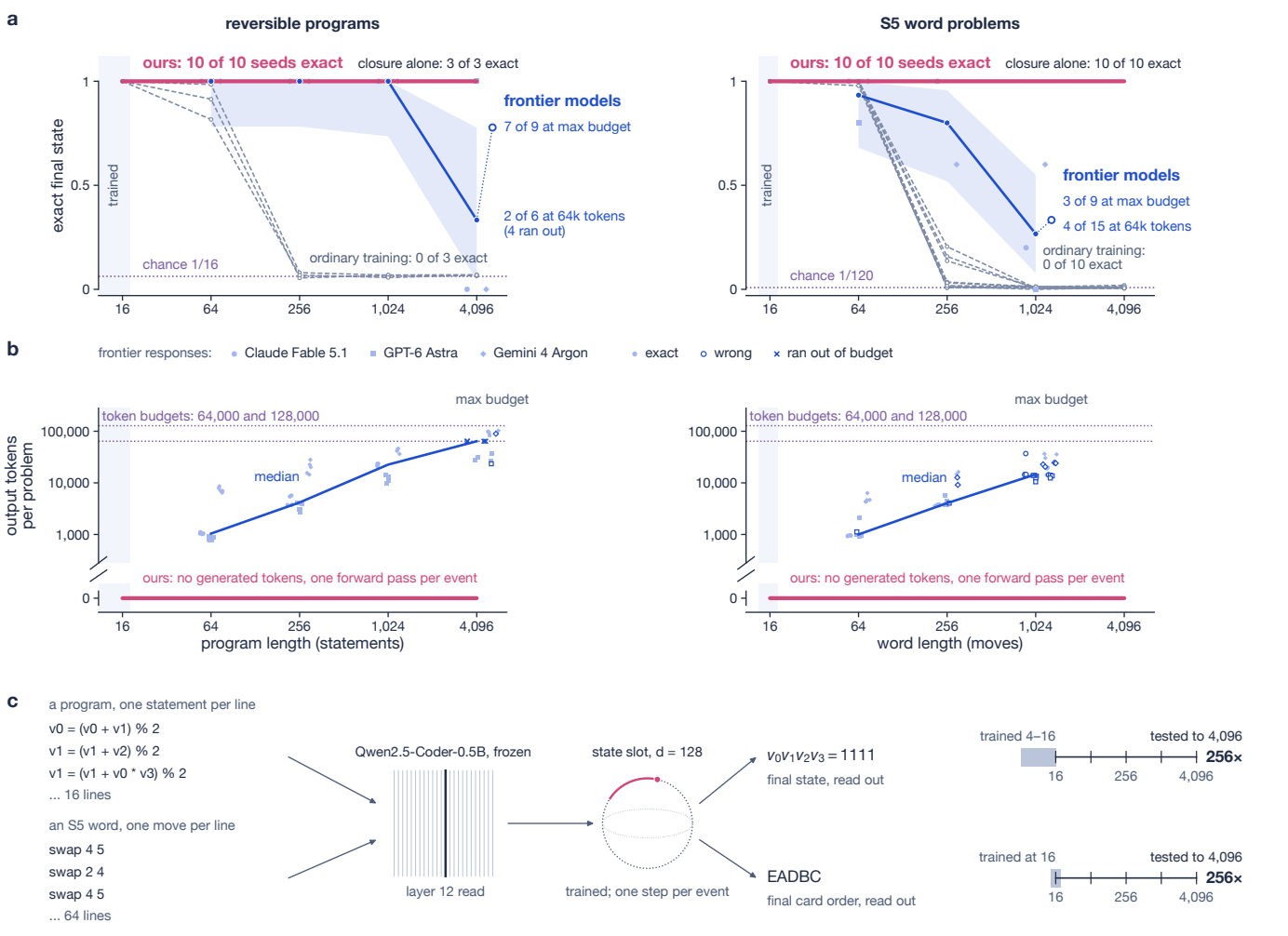
a
reversible programs
ours: 10 of 10 seeds exact
closure alone: 3 of 3 exact
exact final state
trained
frontier models
7 of 9 at max budget
2 of 6 at 64k tokens
(4 ran out)
chance 1/16
ordinary training: 0 of 3 exact
S5 word problems
ours: 10 of 10 seeds exact
closure alone: 10 of 10 exact
frontier models
3 of 9 at max budget
4 of 15 at 64k tokens
ordinary training:
0 of 10 exact
chance 1/120
b
frontier responses:
Claude Fable 5.1
GPT-6 Astra
Gemini 4 Argon
exact
wrong
ran out of budget
max budget
token budgets: 64,000 and 128,000
output tokens
per problem
median
ours: no generated tokens, one forward pass per event
program length (statements)
word length (moves)
c
a program, one statement per line
v0 = (v0 + v1) % 2
v1 = (v1 + v2) % 2
v1 = (v1 + v0 * v3) % 2
... 16 lines
an S5 word, one move per line
swap 4 5
swap 2 4
swap 4 5
... 64 lines
Qwen2.5-Coder-0.5B, frozen
layer 12 read
state slot, d = 128
trained; one step per event
$v_0v_1v_2v_3 = 1111$
final state, read out
EADBC
final card order, read out
trained 4–16
tested to 4,096
256×
trained at 16
tested to 4,096
256×